

Exchanger – CLI Based Second-Hand Marketplace

Course: Programming in Python

Name: MD Alamin

ID: 21-45705-3

1. Introduction

This project is a command-line based second-hand exchange marketplace system developed using Python. The main purpose of this project is to demonstrate core Python programming concepts, Object-Oriented Programming (OOP), and data persistence using JSON files. The system allows users to add products, search products, update and delete products, and find smart exchange matches based on price, category, and condition.

2. Project Objectives

- Implement Object-Oriented Programming using at least four classes.
- Store and load product data using JSON so that data persists after restart.
- Implement full CRUD operations (Create, Read, Update, Delete).
- Implement search, filtering, and sorting features.
- Design a smart exchange matching algorithm.
- Handle invalid user inputs without crashing the program.

3. System Design

3.1 Product Class (Entity)

The Product class represents a single product in the system. It stores product ID, name, category, price, condition, and description.

3.2 ProductManager Class (Business Logic)

The ProductManager class handles all core operations such as adding products, updating price, deleting products, searching products, generating summary reports, and implementing the smart exchange matching algorithm.

3.3 Storage Class (Persistence Layer)

The Storage class is responsible for saving and loading data from a JSON file. This ensures that product data is not lost after program restart.

3.4 CLI Class (User Interface)

The CLI class handles user interaction through a menu-driven interface. It connects user inputs with the ProductManager logic.

4. Smart Exchange Matching Algorithm

The exchange matching system calculates a match score based on three main factors:

- Category similarity (40%)
- Price similarity (40%)
- Condition similarity (20%)

Price tolerance is used to allow exchange within an acceptable range. Products with higher match scores are recommended first.

5. Features Implemented

- Add new product
- View all products
- Search products by keyword
- Update product price
- Delete product
- Smart exchange matching
- Summary report (total products, average price, highest price)
- JSON data persistence
- Input validation handling

6. Conclusion

Through this project, I learned how to design a structured Python application using OOP concepts, file handling with JSON, and implementing algorithms for matching logic. This project improved my understanding of real-world application development. In future, this system can be extended into a web-based platform with user authentication and database integration.